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Cybersecurity & Blockchain

KGSP Enrichment [Week 3 & 4 – 2026]

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للعلوم والتقنية
King Abdullah University of
Science and Technology

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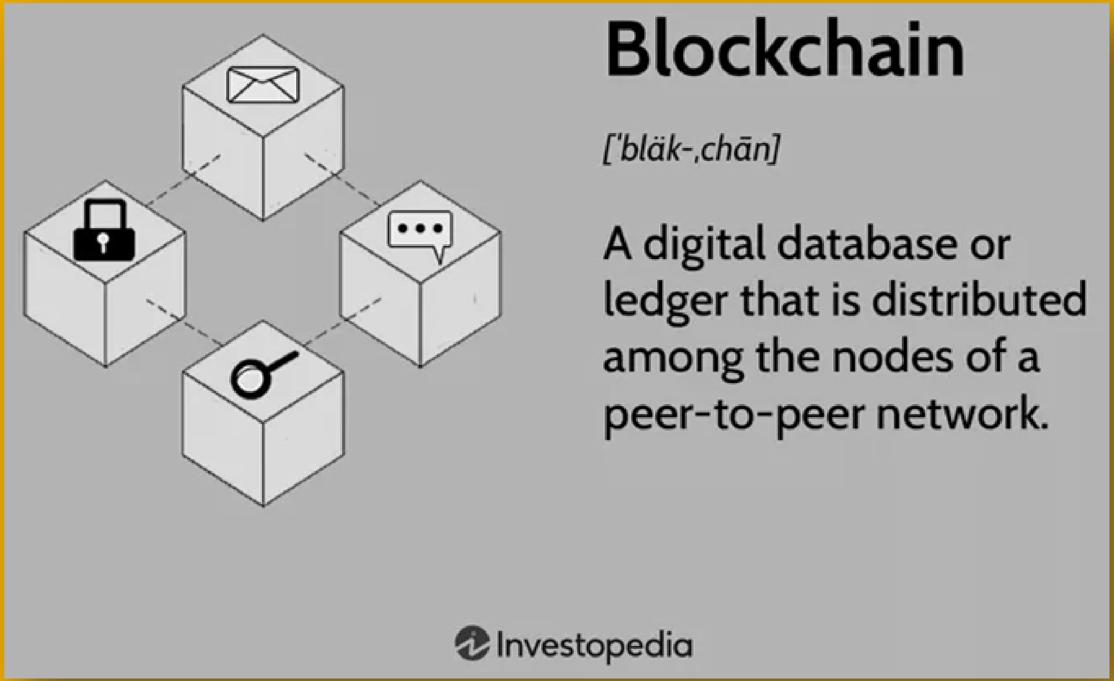
Day 09 – Introduction to Blockchain



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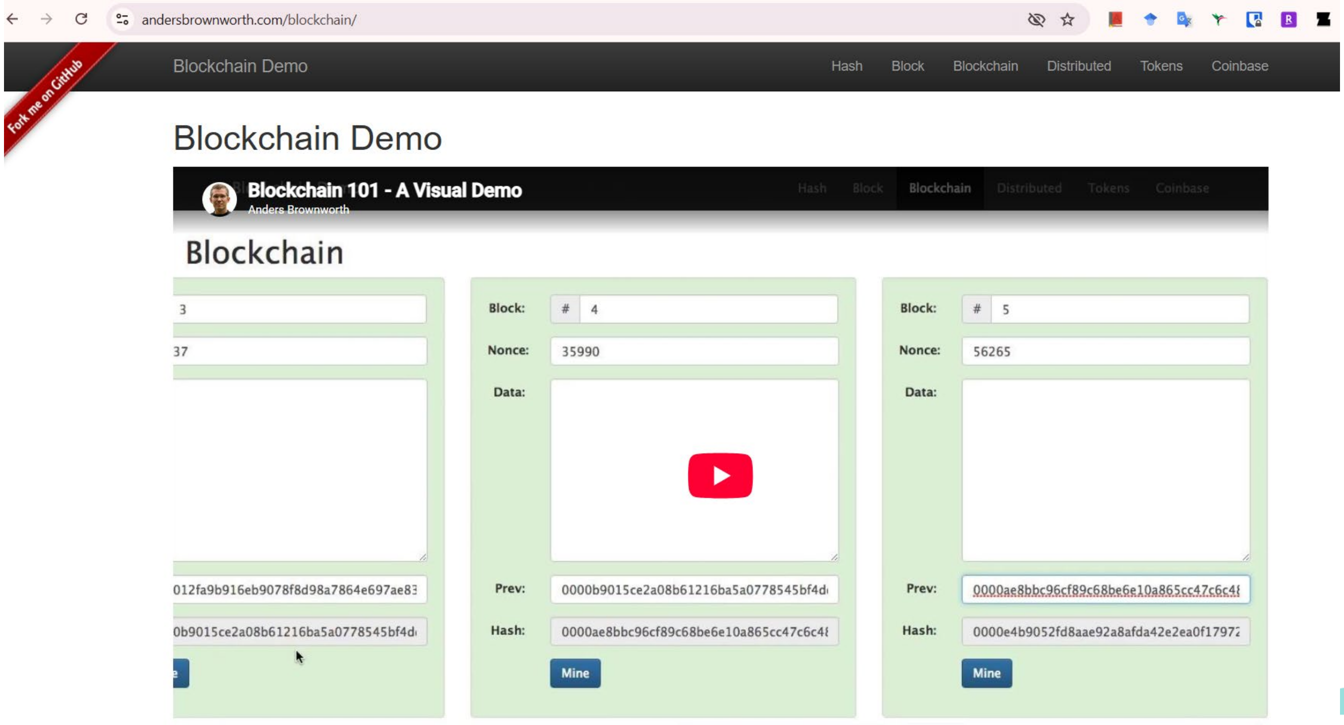
Core Concepts (click image to try)



Foundational Mechanics - I

- Think of a blockchain as a digital ledger of transactions that is shared across a network of computers. Instead of a single database managed by a bank, everyone has an identical copy.
- To understand why this ledger is impossible to cheat, we need to look at how a single **block** is structured.

⛓️ Anatomy of a Block

- Every block in the chain contains three main things:
  - 📝 **The Data:** For example, transaction details (e.g., "Alice sent Bob 5 BTC").
  - 🔑 **Its Own Hash:** A unique digital fingerprint of the block's entire contents, generated by a cryptographic algorithm (usually **SHA-256**).
  - 🔗 **The Previous Block's Hash:** This is the magic link that connects the blocks together.

Revisiting the Power of Hashing

- A cryptographic hash function is like a meat grinder: you put data in, and it spits out a unique, fixed-length string of characters.
  - It is **one-way**: You can easily get the hash from the data, but you can never guess the data from the hash.
  - It is **deterministic**: The exact same data will always produce the exact same hash.
  - It is **avalanche-effected**: If you change even a single character in the original data, the resulting hash changes completely.
- Each block contains the hash of the *previous* block=>
  - changing the data inside block #1 would change block #1's hash
  - This immediately breaks the link to block #2
  - the entire rest of the chain becomes invalid!
- A successful attack will require the attacker to recalculate the hashes for *every single block* that comes after the altered one to make the chain look valid again

Centralized vs Decentralized vs Distributed

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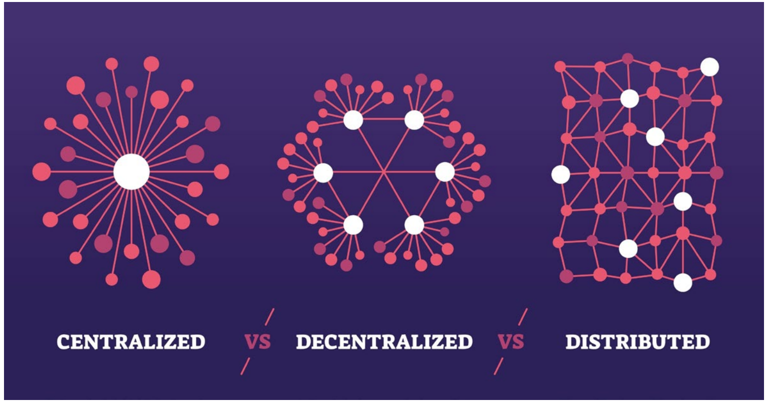
- Instead of one central server holding the database, a blockchain is distributed across thousands of computers (called **nodes**) worldwide. Everyone has an identical copy of the ledger.
- If a hacker successfully tampers with a block and recalculates all the subsequent hashes on *their own* computer, how do you think the rest of the network detects that this altered ledger is invalid?
  - By comparing the final hash on the attacker computer with final hashes on other nodes -> The network compares the hashes, and if one node's ledger doesn't match the rest, it gets rejected.

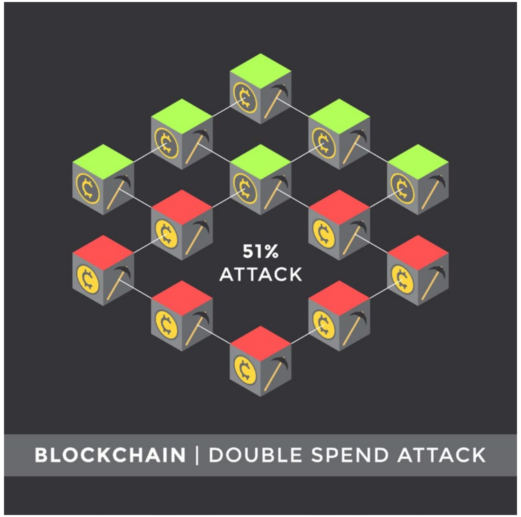


Majority Consensus Algorithm

- Public blockchains rely on a majority consensus (specifically, more than 50%).
- If more than half of the network's voting power agrees on a specific version of the ledger, that version is accepted as the absolute truth. This is why a hacker would need to control at least 51% of the network to successfully alter past transactions—a scenario known as a 51% attack.
- If the network just relies on a "majority vote" of nodes:

what is stopping a hacker from simply spinning up millions of free, virtual nodes on a single computer to easily outvote everyone else?

* Creating fake identities to subvert a network is called a **Sybil Attack**



Secure Consensus Algorithm

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🛑 Voting isn't free: To participate in consensus, you must prove you have committed a scarce resource

💻 Proof of Work (PoW)

- The Puzzle: Miners compete to solve a cryptographic puzzle. They must find a specific number (called a nonce) that, when combined with the block's data, produces a hash starting with a [specific number of zeros](#).
- The Guessing Game: There is no shortcut. Computers must guess trillions of times per second (using electricity).
- The Reward: The first miner to find the correct hash broadcasts it to the network. Once other nodes easily verify it, that miner is "given" the block and rewarded with newly minted cryptocurrency (e.g., Bitcoin) plus transaction fees.

💰 Proof of Stake (PoS)

You must lock up financial capital (cryptocurrency) as collateral.

- The Stake: To participate, you must lock up (stake) a certain amount of the network's native cryptocurrency (e.g., Ether) as collateral.
- The Selection: The protocol uses a semi-random algorithm to select a validator to propose the next block. The more coins you stake, the higher your chances of being chosen (like buying more lottery tickets).
- The Security: Other validators verify the proposed block. If the chosen validator tries to approve fraudulent transactions, the network confiscates a portion of their staked coins (a process called slashing 🔪).

Digital Signatures

- In a blockchain network, your wallet consists of two keys:
    1. 🔑 Private Key: Your secret digital signature. Only you know this. It is used to write a mathematical "signature" on a transaction, proving you authorize the transfer of funds.
    2. 🔓 Public Key: Your public address (like an email address or bank account number). Anyone on the network can see this and use it to verify that the signature was indeed created by your corresponding private key, without ever seeing your private key.

Advantages of Blockchain Technology

- Decentralization: The decentralized nature of blockchain technology eliminates the need for intermediaries, reducing costs and increasing transparency.
- Security: Transactions on a blockchain are secured through cryptography, making them virtually immune to hacking and fraud.
- Transparency: Blockchain technology allows all parties in a transaction to have access to the same information, increasing transparency and reducing the potential for disputes.
- Efficiency: Transactions on a blockchain can be processed quickly and efficiently, reducing the time and cost associated with traditional transactions.
- Trust: The transparent and secure nature of blockchain technology can help to build trust between parties in a transaction.

Disadvantages of Blockchain Technology

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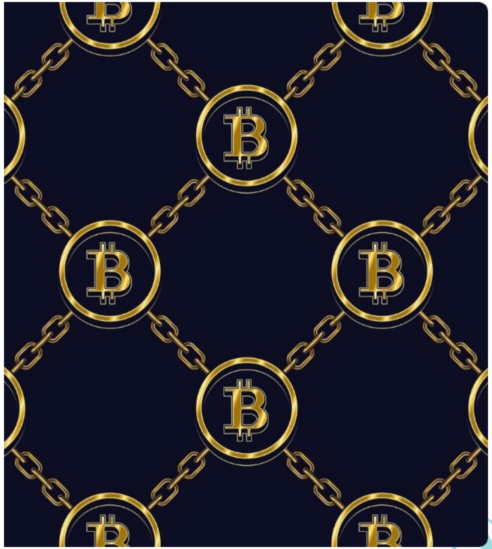
- Scalability: The decentralized nature of blockchain technology can make it difficult to scale for large-scale applications.
- Energy Consumption: The process of mining blockchain transactions requires significant amounts of computing power, which can lead to high energy consumption and environmental concerns.
- Adoption: While the potential applications of blockchain technology are vast, adoption has been slow due to the technical complexity and lack of understanding of the technology.
- Regulation: The regulatory framework around blockchain technology is still in its early stages, which can create uncertainty for businesses and investors.
- Lack of Standards: The lack of standardized protocols and technologies can make it difficult for businesses to integrate blockchain technology into their existing systems.

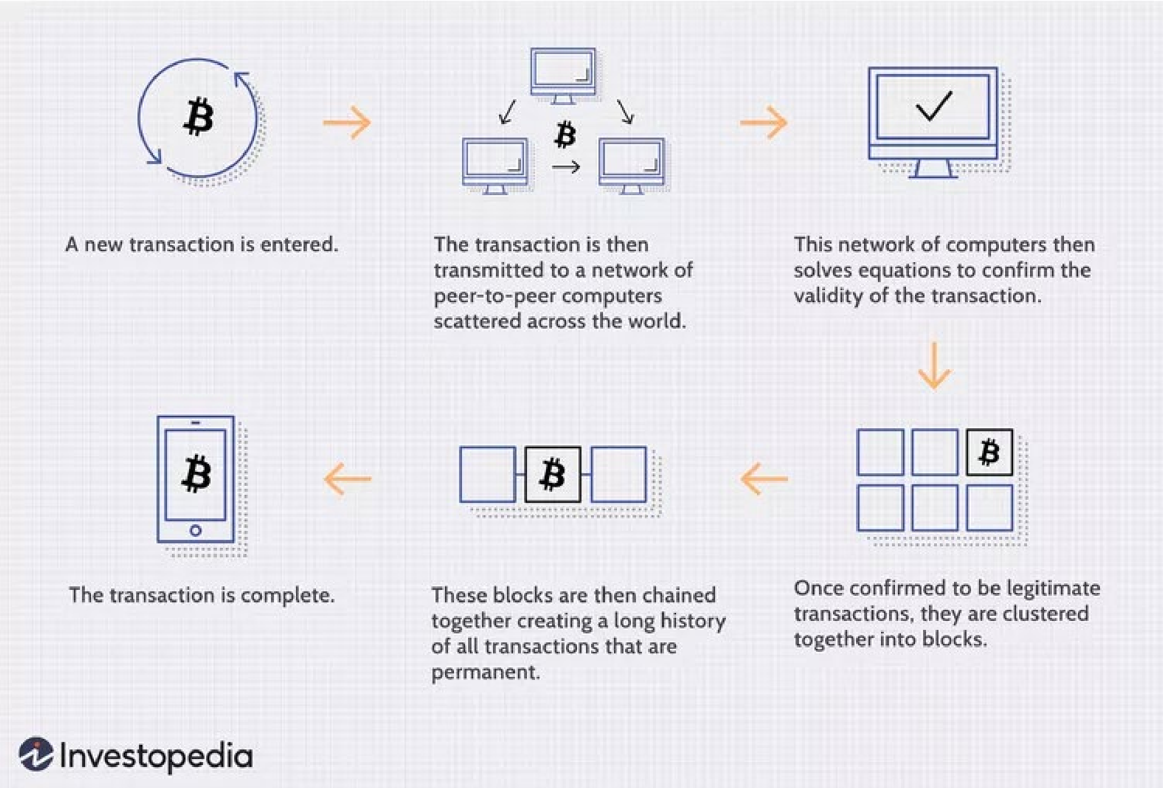
"The advantages of blockchain technology are significant and have the potential to revolutionize many industries. However, there are also several challenges and disadvantages that must be addressed before the technology can reach its full potential."

The Genesis of Bitcoin

- The Enigmatic Creator: Satoshi Nakamoto is the name used by the anonymous individual or group who designed and built the original Bitcoin protocol.
- The moniker first appeared in late 2008 on a cryptography mailing list, introducing a fully peer-to-peer electronic cash system.
- In late 2010, Satoshi handed control of the source code repository to Gavin Andresen and vanished, leaving the project fully decentralized.
- Peer-to-Peer Network
  - Published on October 31, 2008, the whitepaper proposed a system for electronic transactions without relying on centralized trust, using cryptographic proofs to facilitate direct commerce between peers.



How Bitcoin works



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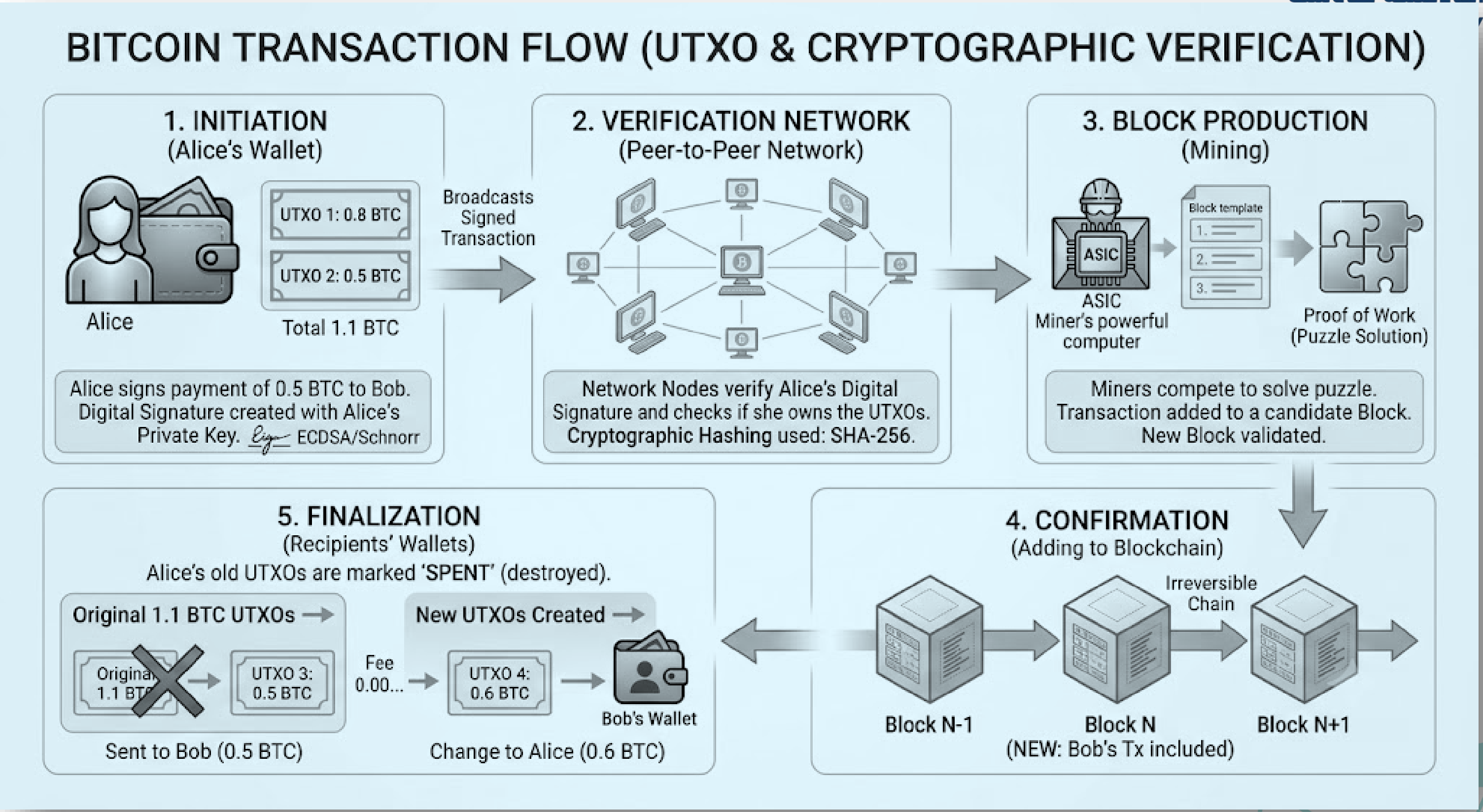

Blockchain beyond Crypto

- Immutable Provenance: Establishes a transparent, unalterable trail of an asset's journey from raw materials to final delivery.
- Rapid Recall Orchestration: Pinpoints contaminated products or defective batches in seconds, saving operational costs.
- Anti-Tampering Control: Integrates IoT devices to log shipping telemetry, validating cold chain requirements securely.